ROUTEOPTIMIZER

Executive Summary

Fleet and route optimization for mining operations - from research prototype to downloadable software mines can run on their own data.

Opportunity

RouteOptimizer is an A-star dispatch simulator and a mixed-integer linear program (MILP) for haul-fleet planning. Today both solvers lock the route before they optimize: the interactive path uses a hard-coded toy mine, and the batch path solves each arc alone. Neither can send a truck a different way when the shift changes, and two known solver defects can drop a better route. The planned work turns this into software a mining company can install, feed with its own haul data, and run on its own machines - with no commercial solver license and no need to send production data off site.

What we will build

Six epics. Epics 1 and 2 stand alone. Epic 3 needs both. Epic 4 needs Epic 3. Epic 5 needs Epics 1 and 2 only. Epic 6 ships on Epics 1, 2, and 5 and does not wait on alternate arcs or replanning.

01 **Keep the better route**
Fix today's solvers: keep a cheaper waiting assignment, tell two arrival times of the same fleet apart, and read real arc endpoints in the linear model.

02 **Run both solvers on a mine you specify**
One scenario schema feeds the dispatcher and the batch job. Load from the toy mine, the fleet form, or haul history. Incomplete data is rejected by name.

03 **Send trucks on different arcs**
Search considers more than one assignment of a truck class. Cost is travel time in minutes. A truck stays on its arc until that time is used up.

04 **Replan from where the trucks are**
Plan the next k segments from the current fleet, including trucks still in transit. Replan when a shovel stops, a queue fills, or demand changes.

05 **Produce one shift plan across the mine**
Replace per-arc LPs with one flow over the whole mine. Conserve trucks, respect queue capacity and fleet size, and take tons and shift length from Parameters.

06 **Download and run it at your mine**
License, documented scenario file format, pip package and CLI, time-limited solves, Docker image, and a Hugging Face repository with samples and install links.

Delivery model

This is optimization software, not a trained machine-learning model. Each mine supplies its scenario and the solvers build a plan locally. Distribution is a pip package and a Docker image; Hugging Face is the front door (README, samples, install links), not a place to host weights. Haul history and tonnage targets stay on the mine's machines.

Units and inputs

Input	Unit	Source
Demand and truck capacity	Tons	Scenario / Parameters
Shift length	Hours	Scenario / Parameters
Arc travel time	Minutes	Scenario arcs / haul history
Locations, arcs, fleet size	Named entities / trucks	Form, fixture, or haul export

Decisions still required

License choice and confirmation of rights with the original authors (credited in the README). An anonymized sample from real haul history publishes only with the data owner's approval; until then the toy mine is the only public sample. Converters for specific fleet-management systems (MineStar, Wenco, and others) are out of scope for this backlog; mines convert into the documented scenario format.